

A KMS_3 state on the Bost–Connes–Marcolli GSp_4 system

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Status: candidate counterexample, likely valid. The construction below gives a normalized scaling measure at inverse temperature $\beta = 3$. By the source paper’s measure–state correspondence, this produces a KMS_3 state and refutes the stated conjecture at one of its two exceptional temperatures.

Abstract

The source paper conjectures that its GSp_4 system has no KMS_β state at $\beta = 1$ or 3 . Its linear-subspace construction indeed cannot produce these exponents. We use instead the nonlinear hypersurface of ordered symplectically orthogonal pairs. At every finite prime, the canonical hypersurface measure is Radon, has finite nonzero mass on the integral locus, and transforms under a symplectic similitude g by $|\lambda(g)|_p^3$. The normalized restricted product therefore transforms under rational g by $\lambda(g)^{-3}$. Combining it with Haar measure on $\text{PGSp}_4^+(\mathbb{R})$ gives exactly the normalized measure required for a KMS_3 state.

1 Claim and source reduction

Let $V = \mathbb{Q}^4$ carry its standard nondegenerate alternating form $\omega(v, w) = v^\top J w$. Write $\lambda(g)$ for the multiplier of $g \in \text{GSp}_4$, so that

$$\omega(gv, gw) = \lambda(g)\omega(v, w), \quad \det g = \lambda(g)^2.$$

Let $\Gamma = \text{Sp}_4(\mathbb{Z})$ and $P = \text{PGSp}_4^+(\mathbb{R})$.

Proposition 2.12 of the source paper identifies KMS_β states with Γ -invariant Radon measures μ on

$$P \times \text{MSp}_4(\mathbb{A}_f)$$

whose induced measure on $\Gamma \backslash (P \times \text{MSp}_4(\widehat{\mathbb{Z}}))$ has mass one and which satisfy

$$\mu(gB) = \lambda(g)^{-\beta} \mu(B) \quad (g \in \text{GSp}_4^+(\mathbb{Q})). \tag{1}$$

The source then conjectures nonexistence at $\beta = 1, 3$; see its Remark 1 on printed page 38.

Theorem 1. *The GSp_4 system of [arXiv:2211.07778](#) admits a KMS_3 state. Consequently, the source conjecture that there are no KMS_β states in the two cases $\beta = 1, 3$ is false.*

The point is that linear subspaces are unnecessarily restrictive. A codimension-one nonlinear cone supplies precisely the missing exponent.

2 The local symplectic hypersurface measure

Fix a prime p and give $\mathbb{Q}_p^4$ additive Haar measure with $\text{vol}(\mathbb{Z}_p^4) = 1$. Put

$$\Sigma_p = \{(v, w) \in \mathbb{Q}_p^4 \times \mathbb{Q}_p^4 : \omega(v, w) = 0\}.$$

For $\Phi \in C_c(\mathbb{Q}_p^4 \times \mathbb{Q}_p^4)$ define

$$\nu_p(\Phi) = \lim_{n \rightarrow \infty} p^n \int \Phi(v, w) \mathbf{1}_{p^n \mathbb{Z}_p}(\omega(v, w)) \, dv \, dw. \quad (2)$$

Lemma 1. *The limit in (2) exists and defines a positive Radon measure supported on Σ_p . Moreover,*

$$\nu_p(\Sigma_p \cap (\mathbb{Z}_p^4)^2) = c_p := \frac{1 - p^{-4}}{1 - p^{-3}}, \quad (3)$$

$$\nu_p(gE) = |\lambda(g)|_p^3 \nu_p(E) \quad (g \in \text{GSp}_4(\mathbb{Q}_p)). \quad (4)$$

The locus where v, w are linearly dependent has ν_p -measure zero.

Proof. For fixed nonzero v , the map $w \mapsto \omega(v, w)$ is a nonzero linear functional. Formula (2) is its Haar disintegration at zero. On compact sets its density is bounded by a constant times $\|v\|_p^{-1}$. This is locally integrable in four variables: on the shells $p^k \mathbb{Z}_p^4 \setminus p^{k+1} \mathbb{Z}_p^4$ the relevant sum is a constant multiple of $\sum_{k \geq 0} p^{-3k}$. Dominated convergence first for compactly supported locally constant functions, followed by the Riesz representation theorem, gives a positive Radon measure. A test function supported away from $\omega = 0$ eventually contributes zero, so the measure is supported on Σ_p .

For the exact integral mass, let $k(v) = \min_i v_p(v_i)$ for $v \in \mathbb{Z}_p^4 \setminus \{0\}$. If $n > k(v)$, then

$$\text{vol}\{w \in \mathbb{Z}_p^4 : \omega(v, w) \in p^n \mathbb{Z}_p\} = p^{-(n-k(v))}.$$

The shell $k(v) = k$ has volume $p^{-4k}(1 - p^{-4})$. Hence

$$c_p = (1 - p^{-4}) \sum_{k \geq 0} p^{-4k} p^k = \frac{1 - p^{-4}}{1 - p^{-3}},$$

which is finite and positive.

For (4), make the change of variables $(v, w) = (gu, gz)$. The ambient Jacobian is $|\det g|_p^2 = |\lambda(g)|_p^4$, while the delta approximation in (2) contributes $|\lambda(g)|_p^{-1}$ because $\omega(gu, gz) = \lambda(g)\omega(u, z)$. Their product is $|\lambda(g)|_p^3$. Finally, for each $v \neq 0$, the dependent vectors $w \in \mathbb{Q}_p v$ form a Haar-null subset of the three-dimensional hyperplane $v^\perp$; integration in v proves the last assertion. $\square$

Embed Σ_p in the symplectic monoid by

$$\iota_p(v, w) = \begin{pmatrix} v & w & 0 & 0 \end{pmatrix}, \quad (5)$$

where the displayed vectors are columns. Since $\omega(v, w) = 0$, one has $\iota_p(v, w)^\top J \iota_p(v, w) = 0$, so the multiplier is zero and $\iota_p(v, w) \in \text{MSp}_4(\mathbb{Q}_p)$. Also $g \iota_p(v, w) = \iota_p(gv, gw)$; thus the support is left GSp_4 -stable. It is nonlinear, and almost every matrix on it has rank two.

3 Restricted product and proof of Theorem 1

Normalize $\bar{\nu}_p = c_p^{-1} \nu_p$. By (3), the compact open set

$$K_p = \Sigma_p \cap (\mathbb{Z}_p^4)^2$$

has $\bar{\nu}_p$ -mass one. The restricted product

$$\bar{\nu}_f = \bigotimes_p' \bar{\nu}_p$$

is therefore a Radon measure on the finite-adelic hypersurface

$$\Sigma(\mathbb{A}_f) = \{(v, w) \in \mathbb{A}_f^4 \times \mathbb{A}_f^4 : \omega(v, w) = 0\}.$$

Push it forward under the adelic version of (5), and denote the resulting measure on $\mathrm{MSp}_4(\mathbb{A}_f)$ by μ_f . It satisfies

$$\mu_f(\mathrm{MSp}_4(\widehat{\mathbb{Z}})) = 1. \quad (6)$$

For $g \in \mathrm{GSp}_4^+(\mathbb{Q})$, only finitely many local factors move the distinguished compact sets. Lemma 1 and the product formula give

$$\mu_f(gE) = \prod_p |\lambda(g)|_p^3 \mu_f(E) = \lambda(g)^{-3} \mu_f(E). \quad (7)$$

The equality is immediate on restricted cylinder sets and extends to Borel sets by regularity.

Let m_∞ be left Haar measure on P , normalized so that its induced measure on $\Gamma \backslash P$ has mass one, and set

$$\mu = m_\infty \otimes \mu_f.$$

This is a Radon measure on $P \times \mathrm{MSp}_4(\mathbb{A}_f)$. It is Γ -invariant: the real factor is left invariant and $\lambda(\gamma) = 1$ for $\gamma \in \Gamma$. A fundamental domain for $\Gamma \backslash P$, crossed with the integral finite-adelic locus, is a fundamental domain for the diagonal action. Equations (6) and the normalization of m_∞ therefore give

$$\mu(\Gamma \backslash (P \times \mathrm{MSp}_4(\widehat{\mathbb{Z}}))) = 1.$$

The real Haar factor is unchanged by rational left translation, while (7) gives

$$\mu(gB) = \lambda(g)^{-3} \mu(B).$$

Thus μ satisfies exactly (1) with $\beta = 3$. Proposition 2.12 of the source paper now produces a KMS₃ state, proving Theorem 1. $\square$

4 Audit, scope, and novelty status

- **No formal delta distribution is assumed.** Equation (2) is a positive limit of ordinary Haar integrals; local integrability at the unique singular point is the convergent $\sum p^{-3k}$ estimate.
- **The exponent is exact.** Two vectors contribute the ambient Jacobian $|\det g|_p^2 = |\lambda(g)|_p^4$ and the single symplectic equation removes one factor $|\lambda(g)|_p$, leaving exponent 3.
- **Normalization is local-to-global.** Dividing by the explicit positive number c_p makes every distinguished integral factor have mass one, so the restricted product is canonical up to the harmless normalization already fixed.

- **Scope.** The packet proves existence at $\beta = 3$ for the GSp_4 system governed by Proposition 2.12. It does not classify all KMS_3 states, decide the separate $\beta = 1$ case, or assert the analogous conclusion for the source’s non-free Connes–Marcolli quotient.

A bounded novelty search on 26 August 2026 checked the four run indexes, the exact conjecture phrase, the paper title, and arXiv searches combining “ GSp_4 ”, “ KMS_3 ”, and “Siegel modular variety”. No later arXiv paper answering this exceptional-temperature conjecture was found. This supports, but does not certify, novelty.

Source

I. Abouamal, *Bost-Connes-Marcolli system for the Siegel modular variety*, arXiv:2211.07778v2 (2023), especially Proposition 2.12 and Remark 1 on printed page 38. <https://arxiv.org/abs/2211.07778>

Remark 1. We have studied the $GS p_4$ -system in the region $\beta > 0$ with $\beta \notin \{1, 2, 3\}$. Let us now consider the cases where the inverse temperature is a pole of the Dirichlet series (41). If $\beta = 2$, it is possible to construct explicit measures $\mu_2 \in \mathcal{E}_2$. We consider the normalized Haar measure on $\mathbb{A}_{f,\mathbb{Q}}$ such that $\text{meas}(\hat{\mathbb{Z}}) = 1$ and

$$\text{meas}(aE) = \prod_p |a_p|_p \text{meas}(E). \quad (55)$$

for any $a \in \mathbb{A}_{f,\mathbb{Q}}^\times$ and measurable subset $E \subseteq \mathbb{A}_{f,\mathbb{Q}}$. Let μ_f be the product measure on $\mathbb{A}_{f,\mathbb{Q}}^2$. Since

$$\begin{pmatrix} 0 & 0 & 0 & 0 & x_1 \\ 0 & 0 & 0 & 0 & x_2 \\ 0 & 0 & 0 & 0 & x_3 \\ 0 & 0 & 0 & 0 & x_4 \end{pmatrix} \in MS p_4(\mathbb{A}_{f,\mathbb{Q}}), \quad x_1, \dots, x_4 \in \mathbb{A}_{f,\mathbb{Q}}$$

we may consider μ_f as a measure on $MS p_4(\mathbb{A}_{f,\mathbb{Q}})$ such that $\mu_f(MS p_4(\hat{\mathbb{Z}})) = 1$. We claim that $\mu_2 = \mu_\infty \times \mu_f \in \mathcal{E}_2$. By construction it is enough to show that μ_2 satisfies the scaling condition (35). Given $g \in GS p_4^+(\mathbb{Q})$, we can find $\gamma_1, \gamma_2 \in \Gamma_2$ and a diagonal matrix $D \in GS p_4^+(\mathbb{Q})$ such that $g = \gamma_1 D \gamma_2$. Since $\gamma \hat{\mathbb{Z}}^4 = \hat{\mathbb{Z}}^4$ for any $\gamma \in \Gamma_2$ and the Haar measure is translation invariant we conclude that $\mu_1(gB) = \lambda^{-2} \mu(B)$ for any Borel subset of $MS p_4(\mathbb{A}_{f,\mathbb{Q}})$.

One would expect to use a similar construction for $\beta = 1$ and $\beta = 3$. However, since the only subspace of $\mathbb{A}_{f,\mathbb{Q}}^4$ stable under the action of $GS p_4(\mathbb{Q})^+$ is $\mathbb{A}_{f,\mathbb{Q}}^4$ itself, this argument fails in the case $\beta = 1$ or $\beta = 3$. We conjecture that the $GS p_4$ -system does not admit any KMS_β state in these two cases.

Figure 1: The source's construction at $\beta = 2$ and its conjecture at $\beta = 1, 3$ (arXiv:2211.07778v2, printed page 38).